

LILaC: Late Interacting in Layered Component Graph for Open-domain Multimodal Multihop Retrieval

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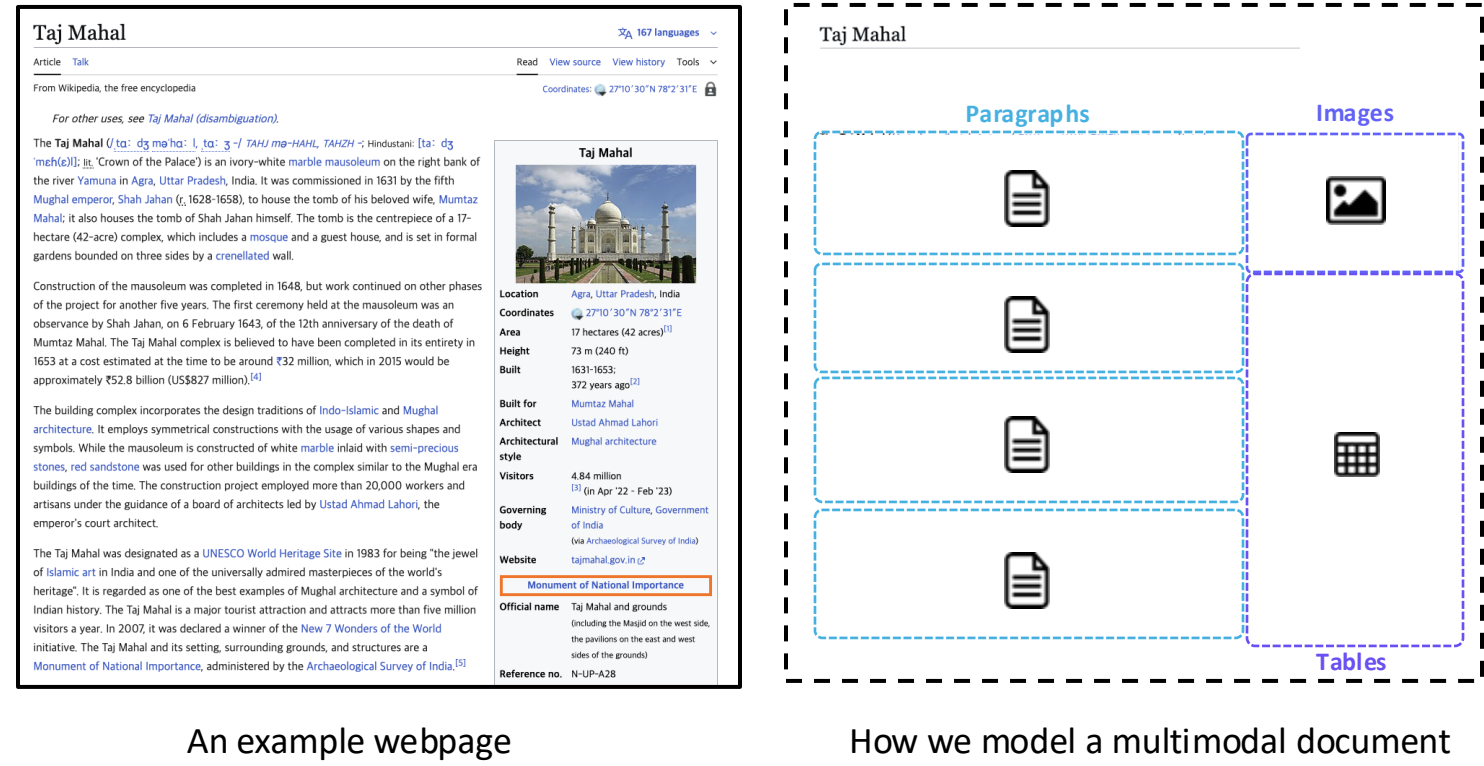
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Problem Description

Multimodal Documents

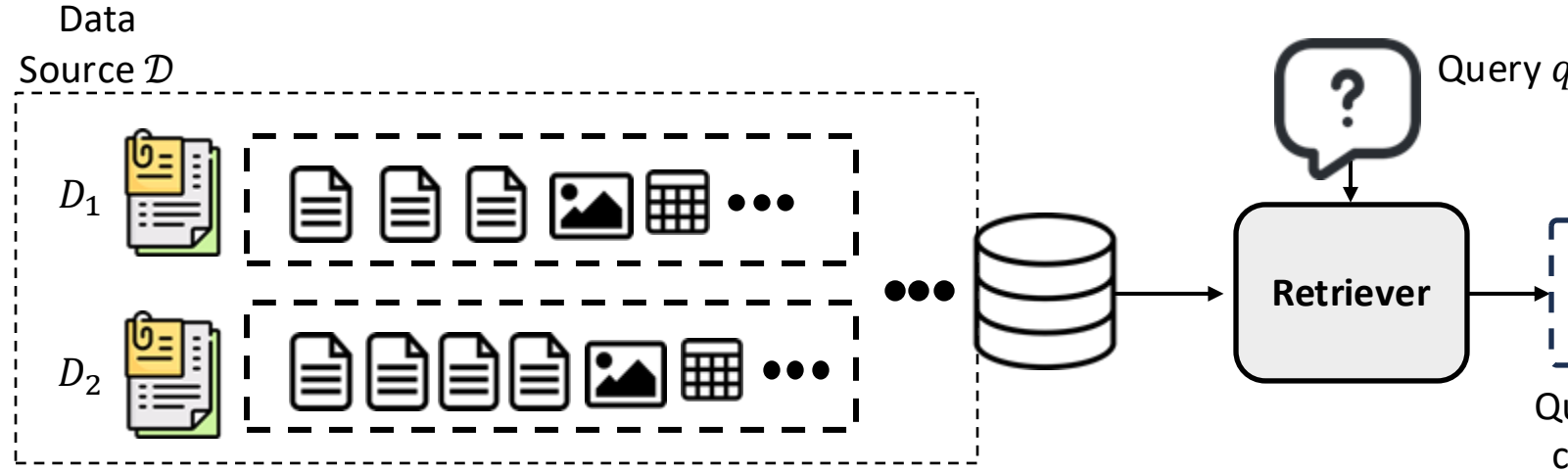
- Real-world documents are often multimodal.
 - e.g., webpages, PDFs, presentation slides, and way more!
- Accurate question-answering involves correctly resolving the relationships between the components.
 - Component:** a multimodal element (i.e., text/table/image)



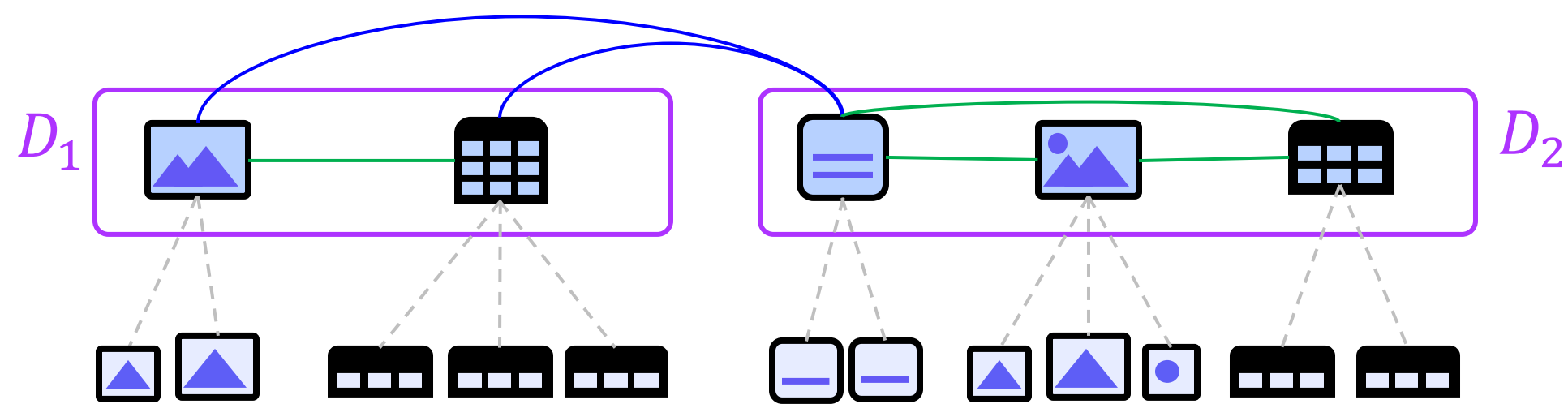
Multimodal Document Retrieval

Given a data source $\mathcal{D}$, a link mapping $\mathcal{L}$ and a query q , correctly retrieve a ranked list of components $\mathcal{R} = [C_1, \dots, C_{n_{ret}}]$. $\mathcal{R}$ should contain the ground truth set of components $\{C_{gt_1}, \dots, C_{gt_r}\}$.

- Data source $\mathcal{D} = \{D_1, \dots, D_{k_{doc}}\}$
- Document $D = [C_1, \dots, C_{k_{comp}}]$
- Component $C \in \{P, T, I\}$
 - P : passage, T : table, I : image
- Link mapping $\mathcal{L} = \mathcal{C} \rightarrow \mathcal{D}$
- $\mathcal{C}$: A set of all components within $\mathcal{D}$

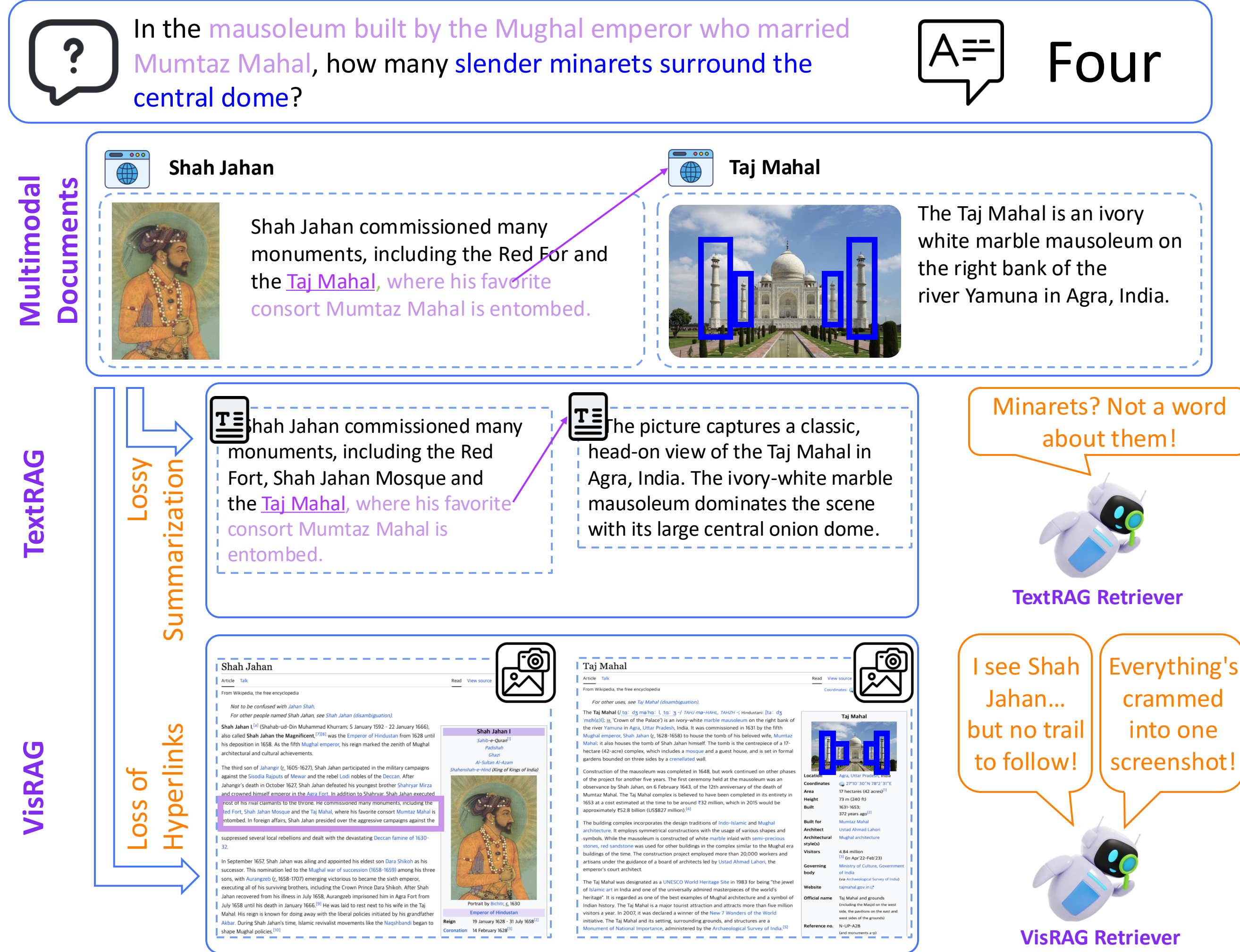


Keys to Accurate Multimodal Document Retrieval

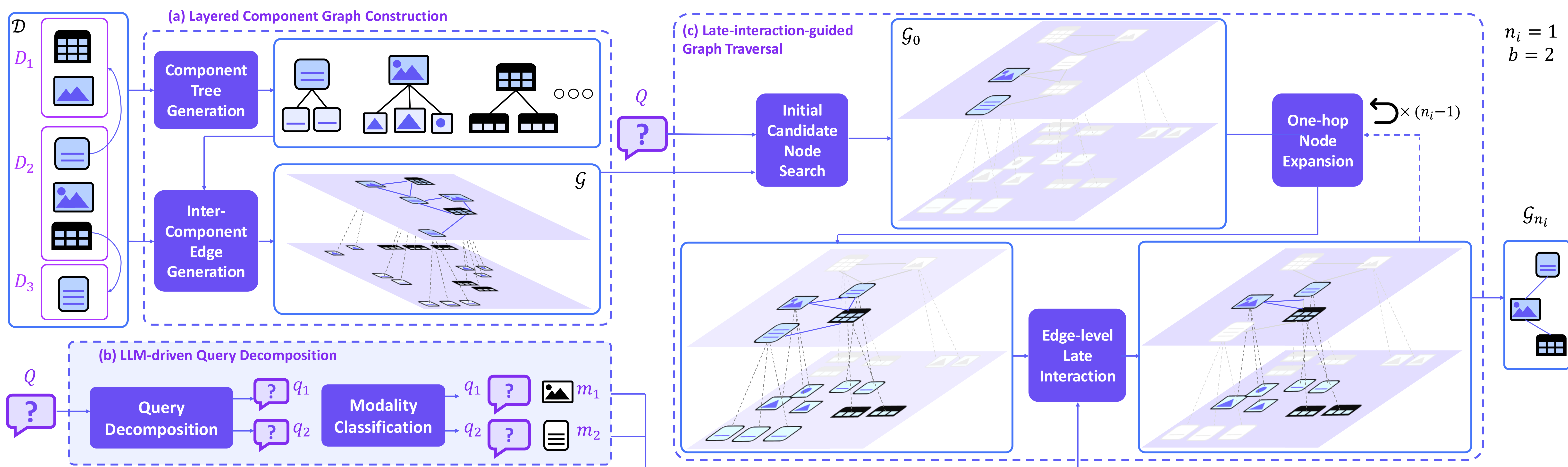


- Modality preservation**
- Consideration of fine-grained granularity:** Paragraph $\rightarrow$ sentences, Table $\rightarrow$ rows, Image $\rightarrow$ detected objects
- Multimodal multihop reasoning:** Exploit both *inter-document links* and *intra-document relationships*

Limitations of Previous Works



The LILaC Algorithm



Brief Explanation about LILaC

Layered Component Graph (LCG)

- Builds a two-layer graph of each document: coarse nodes (paragraphs, tables, images) and fine nodes (sentences, rows, detected images).
- Adds edges for (i) hierarchical containment (coarse $\rightarrow$ fine) and (ii) navigational links (within a document clique + cross-document hyperlinks) to support multihop reasoning.

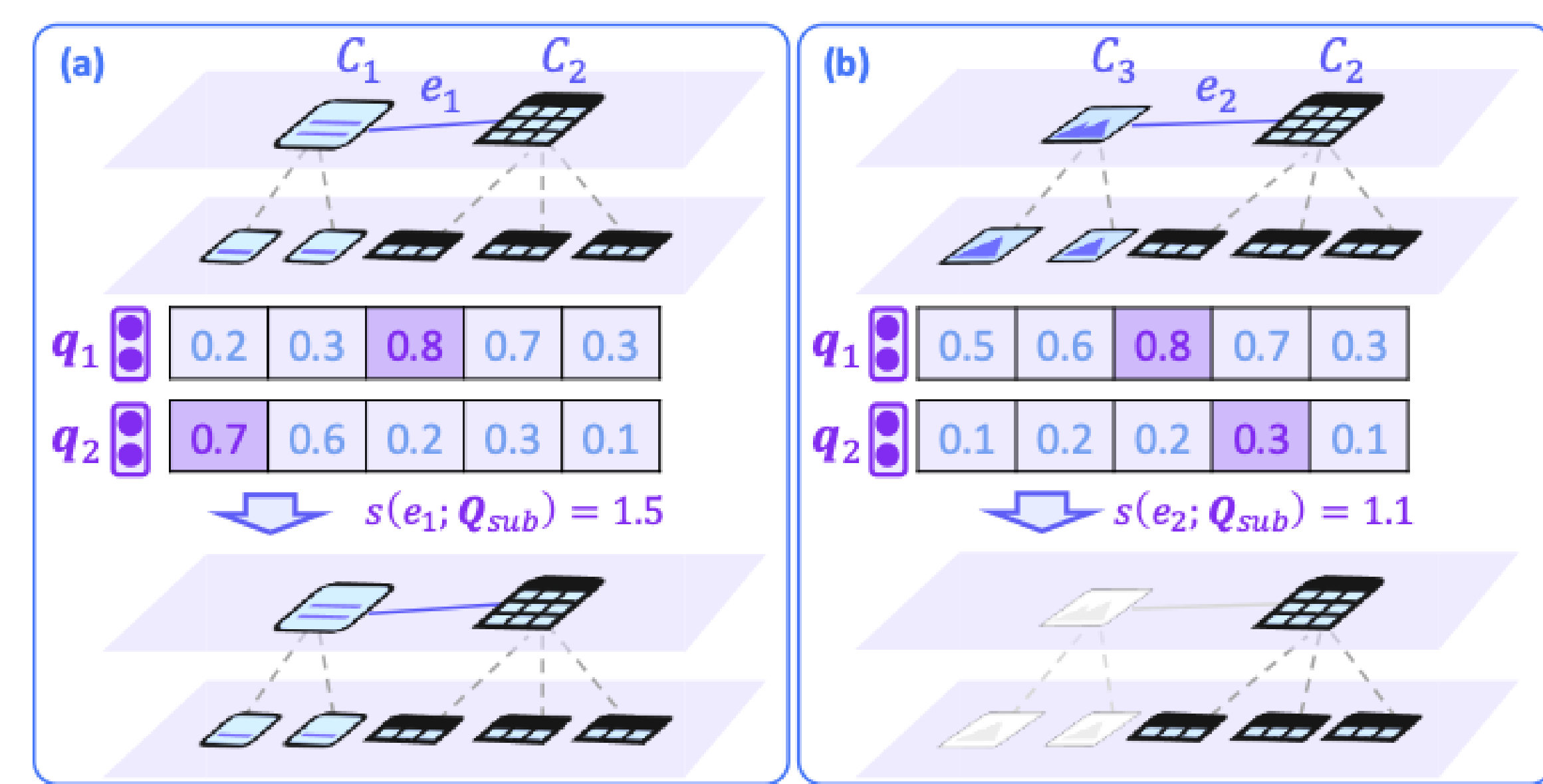
LLM-driven Query Decomposition

- Splits the query into modality-specific subqueries (text/table/image) and embeds each.

Late-Interaction Subgraph Retrieval

- Seed:** conduct a kNN search to find top- b query-relevant components.
- Traverse:** iteratively perform beam search
 - Perform one-hop expansions from an existing subgraph.
 - Leave the top- b query-relevant edges.
 - Late-interaction edge score calculation: $s(e; Q_{sub}) = \sum_{c \in \mathcal{C}} \max_{c \in \mathcal{C}} \text{sim}(c, q)$

Edge-level Late Interaction



Evaluation

Retrieval Accuracy

Algorithm	Embedder Type	MP-DocVQA		SlideVQA		InfoVQA		MultimodalQA		MMCoQA	
		R@3	MRR@10	R@3	MRR@10	R@3	MRR@10	R@3	MRR@10	R@3	MRR@10
NV-Embed-v2	Text	67.85	61.91	88.49	79.55	86.21	80.86	60.19	67.86	46.16	41.45
VisRAG-Ret	Image	83.25	75.55	91.55	84.30	92.76	86.22	50.08	55.08	27.63	23.75
ColPali		80.71	74.86	89.39	81.55	88.30	82.76	58.73	65.05	36.24	32.33
LILaC (w/ mmE5)	Multimodal	61.25	55.30	77.52	68.80	75.09	69.86	54.79	59.02	48.88	40.30
LILaC (w/ UniME)		77.83	71.42	84.35	77.93	82.28	75.27	58.52	61.44	49.63	42.97
LILaC (w/ MM-Embed)		83.59	78.75	92.81	84.43	93.17	86.83	69.07	75.28	55.80	50.77

Question Answering Accuracy

Algorithm	MLLM	MP-DocVQA		SlideVQA		InfoVQA		MultimodalQA		MMCoQA	
		EM	F1	EM	F1	EM	F1	EM	F1	EM	F1
NV-Embed-v2	Qwen2.5-VL 7B	56.51	63.16	53.77	64.41	60.72	63.40	37.23	43.85	28.05	34.67
VisRAG-Ret	MiniCPM V2.6	54.31	68.86	43.88	62.37	50.83	57.55	28.18	34.01	21.51	27.87
VisRAG-Ret	Qwen2.5-VL 7B	65.34	72.24	55.03	66.13	60.16	61.93	22.24	25.55	16.69	20.90
ColPali	Qwen2.5-VL 7B	64.46	71.16	53.77	64.54	58.07	60.38	23.59	27.37	18.07	22.30
LILaC (w/ mmE5)	Qwen2.5-VL 7B	52.96	59.53	50.89	59.07	50.12	53.12	40.72	47.46	33.90	40.38
LILaC (w/ UniME)	Qwen2.5-VL 7B	62.43	69.40	53.05	62.89	53.39	56.86	43.42	49.72	33.39	40.12
LILaC (w/ MM-Embed)	Qwen2.5-VL 7B	65.48	72.42	55.57	66.32	60.91	62.87	44.57	51.97	36.31	43.22

Ablation Study

Embedder Model	Variant	MP-DocVQA		SlideVQA		InfoVQA		MultimodalQA		MMCoQA	
		R@3	MRR@10	R@3	MRR@10	R@3	MRR@10	R@3	MRR@10	R@3	MRR@10
mmE5	LILaC (w/o LCG & QD)	48.90	43.97	75.91	68.13	65.60	58.55	42.99	46.92	41.22	34.51
	LILaC (w/o QD)	60.81	55.02	74.14	67.58	67.70	60.01	45.15	51.12	44.18	36.62
	LILaC	61.25	55.35	76.80	68.99	68.91	61.18	54.78	59.32	48.54	40.22
UniME	LILaC (w/o LCG & QD)	52.12	45.31	81.47	71.22	83.57	77.07	47.68	49.06	45.78	38.41
	LILaC (w/o QD)	77.83	71.27	83.45	75.70	85.11	78.01	52.18	54.01	47.11	39.85
	LILaC	77.83	71.39	84.35	77.93	85.53	78.81	58.43	61.32	49.45	42.91
MM-Embed	LILaC (w/o LCG & QD)	75.80	69.09	92.80	82.19	90.39	83.71	61.10	67.35	47.94	43.75
	LILaC (w/o QD)	82.23	77.75	92.27	83.20	92.17	85.53	63.19	69.91	50.18	45.59
	LILaC	83.59	78.75	92.81	84.43	93.17	86.83	69.07	75.28	55.80	50.77

Runtime Analysis

